

VR AI NOTES

9-Week Project Timeline

Week 1: Foundation & API Integration

02-13-2026

- ☒ ~~Unity project setup~~
 - ☒ ~~XR simulator~~
 - ☒ ~~Basic scene with heart models~~
 - ☒ ~~Set up Groq API (LLM, STT, TTS)~~
 - ☒ ~~Test full pipeline in desktop mode~~
 - ☒ ~~Initial VR build & headset test~~
 - ☐ **Milestone:** AI conversation working, tested in VR
-

Week 2: VR Interaction & Context System

02-20-2026

- ☒ ~~Build and test on Quest headset regularly~~
 - ☒ ~~Implement gaze detection (what user is looking at)~~
 - ☒ ~~Create VisualizationData scriptable object system~~
 - ☒ ~~Add metadata to heart models~~
 - ☐ Update LLM system prompt to be generic (not heart-specific)
 - ☒ ~~Test context-aware responses in VR~~
 - ☒ **Milestone:** Bot knows what you're looking at and responds contextually
-

Week 4: Bot Model & Basic Animation

03-06-2026

- ☐ Source/import humanoid bot model
 - ☐ Position in scene, test scale in VR
 - ☐ Add basic animations (idle, talking gesture)
 - ☐ Test bot visibility and presence in VR
 - ☐ Adjust positioning and scale based on VR comfort
 - ☐ **Milestone:** Bot model present and animated in VR
-

Week 5: Bot Integration & Audio

03-13-2026

- Connect bot animations to audio playback
 - Add simple lip sync or jaw movement
 - Implement bot gaze (looks at user or at visualization)
 - Test audio spatialization in VR
 - Refine bot behavior based on VR testing
 - **Milestone:** Bot speaks with synchronized animation
-

Week 5 & 6: Particle System

04-03-2026

- ☒ ~~Design particle swarm~~
 - ☒ ~~Implement responsive particle behavior~~
 - ☒ ~~Particles react to AI state (idle, listening, speaking)~~
 - ☐ Position and tune for VR visibility
 - ☐ **Milestone:** Both embodiments functional in VR
-

Week 7: Scene Polish & Lighting

04-10-2026

- Finalize scene layout based on VR feedback
 - Implement proper lighting
 - Add environment details (pedestals, subtle UI)
 - Performance optimization for Quest (maintain fps)
 - Test with different users for comfort
 - **Milestone:** Polished, well-lit, comfortable VR scene
-

Week 8: Multiple Visualization Types

04-17-2026

- ☐ Add a simple **bar graph** visualization (cell type counts)
 - ☐ Add metadata to graph
 - ☐ Test bot answering questions about graphs vs 3D models
 - ☐ Refine context detection (handle multiple objects in view)
 - ☐ Continue VR testing and comfort optimization
 - ☐ **Milestone:** Bot works with different visualization types
-

Week 9: Final Polish & Documentation

04-24-2026

- Final testing with multiple visualization types
 - Record demo video (both embodiments)
 - Update documentation
 - **Milestone:** Complete, demo-ready, documented
-

10/02/2026

- Scene
 - Basic scene with human heart. Could ask the bot to tell how two heart models differ or maybe which cell type occurs the most?
 - Use HRA 2.4 models for it.
<https://github.com/cns-iu/hra-organ-gallery-in-vr-development/tree/main/hra-organ-gallery/Assets/Models/Organs/2.4/HiRes>
 - No constraints on scene wrt the skybox, env, etc. There's flexibility.
- Bot
 - Humanoid
 - Humanoid model - stay away from the uncanny valley.
 - Look at Fab, Unity assets, Sketcfab.
 - Could import animations from Mixamo, the model needs to be properly rigged for this.
 - If not mixamo, use Blender to create animations.

Particles

- Use a particle system. Something like Navi from Legend of Zelda
<https://youtube.com/shorts/ShXbpyPgbOg?si=ExUsNCbHkE6o5rOU>
- An entity that follows you around (not in an annoying way)
- Maybe a pulsing swarm instead of something that fades over time so that the user does not feel like they have limited time to interact with the entity.
- Andi mentioned HRA red fading to white (color over lifetime)

12/02/2026

Pricing:

Complete AI Pipeline:

1. **STT (Speech-to-Text)**: User talks in VR → converts to text
2. **LLM (Language Model)**: Text question → generates text answer
3. **TTS (Text-to-Speech)**: Text answer → audio for bot to speak

Option A: OpenAI Ecosystem (simplest)

- **STT**: OpenAI Whisper API - \$0.006 per minute
- **LLM**: GPT-4o mini - very cheap (~\$0.15 per 1M input tokens)
- **TTS**: OpenAI TTS - \$15 per 1M characters
- **Pros**: Single API provider, great documentation, reliable
- **Estimated cost for 7-8 weeks**: \$10-30 total

Option B: Mix & Match (cheapest)

- **STT**: OpenAI Whisper - \$0.006/min
- **LLM**: GroqAI (LLaMa 3) - FREE tier is generous, super fast
- **TTS**: ElevenLabs free tier or \$5/month
- **Pros**: Cheapest, GroqAI is extremely fast
- **Estimated cost**: \$0-15 total
- **Cons**: Managing multiple APIs

ElevenLabs has a built-in Unity SDK, and is explicitly marketed for VR.

13/02/2026

- Will have to look into lighting. The models need to be well lit from all angles.
- Create a timeline for the project. We have 9 weeks. April 24th.
- The bot can have basic animations rn (idle and talk)
- This could be a scene in HRA. The idea is for the user to be able to have a conversation about the visualizations and models, and have their queries answered in real time.

17/02/2026

- Groq API key:

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

Orpheus (via Groq Cloud) is a fast, high-quality open-source text-to-speech model. It is designed for real-time applications, often achieving under 200ms time-to-first-byte when powered by Groq's specialized Language Processing Units (LPUs). **ElevenLabs** is a proprietary platform known for its incredibly natural, emotionally expressive voices, and extensive voice cloning capabilities. [🔗](#)

At-a-Glance Comparison

Feature 🔗	Orpheus (Groq)	ElevenLabs
Primary Strength	Ultra-low latency (Speed)	High-fidelity voice/emotion
Speed	Sub-200ms, specialized for RT	Very fast, but typically higher latency
Model Type	Open-source (Canopy Labs)	Proprietary
Voice Quality	Very high, natural (Arabic/English)	Industry-leading, "too perfect"
Best For	Real-time agents, low-cost prod	Content creation, high-end cloning

Key Differences

1. Latency and Speed (Groq Advantage)

Groq's infrastructure is designed specifically to reduce latency for AI models. When using Orpheus, the "time-to-first-byte" can be under 200ms, which is critical for natural, interactive voice conversations. This makes it a strong contender for "real-time" voice agents. [🔗](#)

- Groq has all 3 services (STT, LLM, TTS)
 - STT: Whisper Large v3 Turbo
 - LLM: Llama 4 Scout
 - TTS: Orpheus English

Will try using Orpheus (has fewer voices but very low latency)
- Context for bot:
 - Scriptable Objects that can be attached to each visualization. Can use JSON/text files

that provide metadata from external files (useful for non-programmers but SO better imo)

- The bot can look at a game object and detect through raycast and then pull up context about that specific game object. Makes it more dynamic that way.

19/02/2026

- Used up 3.5k tokens for orpheus tts. The limit is 3.6k. That makes it expensive.
- Elevenlabs Flash on the other hand will give me about 20 minutes of audio where one minute of audio costs approximately 1000 characters.

Compare plans

<

Text to Speech

Speech to Text

Agents

Voice Changer

Sound Effects

Voice Isolator

Music

Image & Video

Audio >

	Free	Starter	Creator	Pro	Scale	Business
Multilingual v2/v3						
Minutes included ⓘ	~10	~30	~100	~500	~2,000	~11,000
Extra minutes ⓘ	×	×	~\$0.30/minute	~\$0.24/minute	~\$0.18/minute	~\$0.12/minute
Audio quality ⓘ	128 kbps, 44.1kHz	128 kbps, 44.1kHz	128 & 192 kbps (via API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz
Flash						
Minutes included ⓘ	~20	~60	~200	~1,000	~4,000	~22,000
Extra minutes ⓘ	×	×	~\$0.15/minute	~\$0.12/minute	~\$0.09/minute	~\$0.06/minute
Audio quality ⓘ	128 kbps, 44.1kHz	128 kbps, 44.1kHz	128 & 192 kbps (via API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz	128 & 192 kbps (via Studio & API), 44.1kHz

- The starter tier is \$5 month and increases usage limit quite a bit.

20/02/2026

- ElevenLabs API key:

Sk_c4fe0a87504049908eb3272b634071d92551b29621d3c450

- Super low latency: ~75ms
- 40k character limit (Orpheus has 3.6k). Could use the free tier for now and see if we need the \$5/month in the future.

24/02/2026

- Try multiple visualization types in the scene and see how that compares to the 3d models and how the AI Manager interacts with those as well.
- Updated Unity version. Using Unity 6.3 LTS (6000.9f1) now.
- Updates some of the deprecated assets and components that I used in the 2022 version.
- Using XR Interaction Simulator instead of XR Device Simulator now since that's Legacy and lacks many of the newer features.

26/02/2026

- Downloaded free 3d models from SketchFab (.blend or .fbx). Not sure if they're rigged.

Updates for today (March 06, 2026):

Still working on the particle swarm. Yesterday I spent some time going through YouTube tutorials on Blender animations and general tools to get more comfortable with the working. The animations are taking a bit longer than expected, but I'm making progress and have a clearer idea of what I need to do.

Will keep pushing on it today.

10/03/2026

- **Type 2 — Orb/Sphere with implied structure.** The white smoky sphere and the blue cellular/energy orb both have a **loosely spherical form** that feels contained but turbulent inside. The blue one especially is interesting because it actually looks like a **cell** — which is a fantastic happy accident given your anatomical context. It has that organic membrane quality with an energetic core.

- This direction has a clear **center of gravity** — the entity "lives" in a defined space, which makes it easier for users to know where to look and talk to. It feels more like a *being* while still being abstract.
- ---
- **My honest take combining both types:**
- I'd actually suggest a **hybrid** — a loose orb as the resting/idle state that occasionally trails off into ribbons or wisps when active or speaking. The sphere gives it a home base, the flowing tendrils give it expressiveness.
- The blue cellular orb is particularly compelling for your project because it visually rhymes with the anatomical content users are exploring.

So the concept would be something like:

Idle state — a gently pulsing orb, particles slowly orbiting/churning within the sphere boundary, calm blue-purple glow

Listening state — orb tightens slightly, maybe a subtle inward pull, indicating attention

Speaking/active state — tendrils start trailing off from the orb, ribbons of smoke/energy flowing outward, more chaotic and expressive, brighter

Thinking/processing — could do a fast swirl or a brief contraction before expanding again

This gives users clear **readable states** which is important for UX in VR — they need to know when to talk and when to wait.

When adding to GITHUB add to ARCHIVES folder

